

EEC ML — All AI Features

#	Feature	Status	Progress %	In Old CSV?	Files / Location	What's Missing
— 1. CORE AI PIPELINE —						
1	Document Ingestion (OCR → Parse → Chunk → Embed → Qdrant)	✓ Done	90%	Yes	ai-service/app/modules/documents/	Learning Outcomes + Bloom not auto-generated on ingest
2	AI Orchestrator (central workflow coordinator)	✗ Not Started	0%	No	NEEDED: ai-service/app/orchestrator/	Node calls endpoints directly — biggest arch violation
3	RAG Engine (semantic retrieval with metadata filtering)	✓ Done	90%	Yes	ai-service/app/modules/retrieval/	Hybrid BM25+semantic not yet done
4	Hybrid Search (BM25 keyword + semantic vector)	✗ Not Started	0%	No	NEEDED in retrieval/service.py	Only pure vector search today
5	Embedding Engine (nomic-embed-text via Ollama)	✓ Done	95%	No	ai-service/app/modules/embeddings/	Not listed as standalone in old CSV
6	Prompt Library (externalised /prompts/ directory)	✗ Not Started	0%	No	NEEDED: ai-service/prompts/	All prompts hardcoded — architecture violation
— 2. CONTENT INTELLIGENCE —						
7	Curriculum Graph (topic → prerequisite map)	◆ Partial	30%	Yes	backend/models/CurriculumMap.js + curriculumMapRoutes.js	No traversal logic, no auto-update on ingest
8	Learning Outcomes Auto-generation from documents	✗ Not Started	0%	No	NEEDED in documents/service.py	Not in ingestion pipeline at all
9	Bloom Engine (tag documents + questions + answers)	✗ Not Started	0%	No	NEEDED: ai-service/app/modules/bloom/	Zero code exists; doc claims ☒ Strong (wrong)
10	Document Versioning (retain old versions, never auto-delete)	✗ Not Started	0%	No	NEEDED in TeachingMaterial.js	No version field on model
11	Question Bank (permanent store, teacher-editable)	◆ Partial	50%	Yes	backend/models/ExamQuestion.js + PracticeQuestion.js	No Bloom level, not AI-auto-saved, no difficulty calibration
— 3. ASSESSMENT & EVALUATION —						
12	Academic Assessment Engine (MCQ + written answer eval)	◆ Partial	40%	Yes	ai-service/app/modules/assessment/	Only language assessment wired; academic MCQ/written answer not
13	Language Reading Assessment (Qwen3 8B scoring)	✓ Done	90%	Yes	ai-service + ReadingAssessment.js + ReadingPracticePage.jsx	Teacher review dashboard polish needed
14	Language Writing Assessment (Qwen3 8B scoring)	✓ Done	90%	Yes	ai-service + WritingAssessment.js + WritingPracticePage.jsx	Teacher review dashboard polish needed
15	Answer Evaluator — full rubric (marks/rubric/Bloom/confidence/)	◆ Partial	40%	No	ai-service/app/modules/assessment/service.py	Only language eval complete; 6 of 10 planned fields missing
16	Error Classification Engine (Concept/Calculation/Reading/Logic)	✗ Not Started	5%	Yes	NEEDED: ai-service/app/modules/error_classifier/	Blocks gap detection and intervention logic
17	Baseline Quiz (initial mastery estimate)	✓ Done	85%	Yes	backend/models/BaselineQuiz.js + BaselineQuiz.jsx	Results not feeding Mastery Engine pipeline
— 4. MASTERY & MEMORY —						
18	Mastery Engine (score calculate + auto-update after every attempt)	◆ Partial	60%	Yes	backend/services/masteryEngine.js + MasteryScore.js	Auto-update not wired from quiz/test results; badge + no decay
19	ML Engine (weighted EMA mastery, at-risk detection, class trends)	◆ Partial	50%	No	backend/services/mlEngine.js + mlRoutes.js	Service exists with EMA decay; at-risk endpoint live; NOT in old CSV
20	Engagement Scorer (time × views × attempts per topic)	◆ Partial	50%	No	backend/services/engagementScorer.js + engagementRoutes.js	Service built; low-engagement swap suggestions; NOT in old CSV
21	Spaced Repetition Engine (SM-2 algorithm, review scheduling)	◆ Partial	55%	No	backend/models/SpacedRepetitionSchedule.js + spacedRepetitionRoutes.js	SM-2 intervals built; not triggered from quiz results automatically
22	Student Academic Memory (weak topics, past mistakes, study history)	✗ Not Started	5%	No	NEEDED: extend StudentMemorySummary.js	Language memory done; academic memory not built
23	Conversation Memory (persisted cross-session)	✗ Not Started	5%	No	TutorConversation.js exists; no cross-session recall	In-memory per request only
— 5. GAP DETECTION & LEARNING PATH —						
24	Gap Detection Engine (traverse graph → root-cause weak topics)	✗ Not Started	5%	Yes	NEEDED: ai-service/app/modules/gap_detector/	Biggest missing engine; doc claims ☒ Strong (wrong)
25	Recommendation Engine (next topic + material based on mastery)	◆ Partial	40%	No	backend/routes/recommendationRoutes.js	Route built using mastery + spaced rep; NOT in old CSV; full algo not
26	AI-generated Learning Path (bridge from gap → target topic)	✗ Not Started	0%	Yes	NEEDED in ai-service; TeacherLearningPath.js exists for manual	Manual paths only; no AI bridge path generation
27	Personalised Study Plan (week-by-week schedule output)	✗ Not Started	0%	Yes	NEEDED: integrate gap → path → study plan generator	No plan generation or calendar output
— 6. AI CONTENT GENERATION (Student) —						
28	AI Tutor Chat — Explain Mode	✓ Done	90%	No	ai-service/app/modules/chat/ (mode=explain)	Mode works; not listed separately in old CSV
29	AI Tutor Chat — Homework Help (Socratic)	✓ Done	95%	Yes	ai-service/app/modules/chat/ (mode=homework_help)	7-rule Socratic enforcement active
30	AI Tutor Chat — Summarize Mode	✓ Done	90%	Yes	ai-service/app/modules/summaries/	Works well
31	AI Tutor Chat — Notes Mode	✓ Done	90%	Yes	ai-service/app/modules/chat/ (mode=notes)	No save/export to student profile
32	AI Tutor Chat — Quiz Mode	✓ Done	90%	Yes	ai-service/app/modules/quiz/ + QuizUI	Results not auto-saved to Question Bank
33	AI Tutor Chat — Flashcard Mode	✓ Done	90%	Yes	ai-service/app/modules/flashcards/ + FlashcardUI	Ratings not persisted to Mastery Engine
34	AI Tutor Chat — Mind Map Mode	✓ Done	90%	Yes	ai-service/app/modules/chat/ (mode=mind_map) + MindMapUI	No export (PDF/image)
35	Worksheet / Homework Generation (personalised to mastery gap)	✗ Not Started	0%	No	NEEDED in ai-service chat module	No personalised worksheet output
36	Adaptive Difficulty (adjust question hardness based on mastery)	✗ Not Started	0%	No	NEEDED: wire MasteryScore into quiz/question generation	All content fixed difficulty today
— 7. AI CONTENT GENERATION (Teacher) —						
37	AI Lesson Content Generator (teacher-facing)	◆ Partial	45%	No	backend/routes/aiTeacherRoutes.js (mode=lesson-content)	Route exists calling /generate/teacher; NOT in old CSV
38	AI Hinge Question Generator (teacher-facing)	◆ Partial	45%	No	backend/routes/aiTeacherRoutes.js (mode=hinge-questions)	Route exists; NOT in old CSV
39	AI Question Paper Generator (teacher selects topic + Bloom)	✗ Not Started	0%	No	mockExamRoutes.js generates questions; no Bloom param	No Bloom-level or difficulty selection
40	Teacher AI Override (correct/edit an AI answer)	✗ Not Started	0%	No	NEEDED: teacher annotation layer	No mechanism for teacher to flag or override AI
— 8. LANGUAGE ASSESSMENT —						
41	Speech Transcription (faster-whisper)	✓ Done	85%	Yes	ai-service/app/modules/speech/service.py	Latency on slow hardware untested
42	Pronunciation Scoring (SpeechBrain)	✓ Done	85%	Yes	ai-service/app/modules/speech/pronunciation.py	Real-device mic testing needed
43	Student Language Profile (adaptive Qdrant memory)	✓ Done	80%	Yes	StudentLanguageProfile.js + language_memory module	Not surfaced as student dashboard card
— 9. ANALYTICS & INTELLIGENCE —						
44	Teacher Analytics Dashboard (class performance)	◆ Partial	40%	Yes	teacherAnalyticsRoutes.js	No per-topic heatmap; no gap insights; no at-risk flag UI
45	Admin Analytics Dashboard (school-level AI metrics)	◆ Partial	35%	Yes	adminAnalyticsRoutes.js	No AI-generated insights; no subject weak-area report
46	Student Progress Dashboard (unified mastery + gaps + path)	◆ Partial	45%	Yes	MasteryView.jsx + LearningPathMapView.jsx	All views independent; no unified health card
47	At-risk Student Detection (ML engine flag)	◆ Partial	40%	No	backend/services/mlEngine.js (GET /class/at-risk)	Logic built; not surfaced to teacher in UI; NOT in old CSV
48	Class Engagement Analytics (per-topic engagement scores)	◆ Partial	40%	No	backend/services/engagementScorer.js + engagementRoutes.js	Service built; no teacher dashboard card; NOT in old CSV
49	AI-generated Insights for Teacher ('Class struggles with X')	✗ Not Started	0%	No	NEEDED: LLM summarise analytics data	No AI narrative generation over analytics
50	Wellbeing Tracking (student check-in + admin visibility)	◆ Partial	40%	No	backend/models/Wellbeing.js + wellbeingRoute.js	Model + CRUD route exist; no safeguard/escalation layer; no UI
— 10. RESEARCH-GRADE FEATURES (from Area to Cover doc) —						
51	Learner Confidence Tracking (dynamic self-perception)	✗ Not Started	0%	No	NEEDED: new model + AI check-in	No model, no UI, no data collection
52	Help-Seeking Behaviour (log when/how student asks for help)	✗ Not Started	0%	No	NEEDED: event logging in tutor session	No tracking mechanism
53	Retention / Forgetting Model (knowledge decay over time)	◆ Partial	40%	No	SpacedRepetitionSchedule.js + spacedRepetitionRoutes.js	SM-2 built; incorrectly listed as 'Future ML' in old CSV
54	Misconceptions Engine (explicit error model per topic/student)	✗ Not Started	0%	No	NEEDED: extend error classifier output	Error classifier itself not built yet
55	Multidimensional Engagement (situational + behavioural + emotional)	◆ Partial	25%	No	engagementScorer.js (behavioural only)	Only behavioural dimension built; situational + emotional not
56	Intervention Effectiveness (measure what actually improves learning)	✗ Not Started	0%	No	NEEDED: analytics overlay on interventions	InterventionLog.js exists; no measurement framework
57	Teacher Escalation (auto-flag students for human support)	✗ Not Started	0%	No	NEEDED: alert system from gap detection / at-risk engine	No alerting or escalation workflow
58	Student Agency (learner control over learning path)	✗ Not Started	0%	No	NEEDED: student path selection UI + preference store	All paths teacher-assigned
59	Explainable AI (why did AI recommend / answer this?)	✗ Not Started	0%	No	NEEDED: reasoning trace in tutor response	Black box currently
60	Social / Belonging Dimension (peer signals + community)	✗ Not Started	0%	No	Alcove exists for posts; no belonging analytics	No social learning signals
61	Evidence Testing (prove platform improves learning outcomes)	✗ Not Started	0%	No	NEEDED: pre/post measurement framework	No outcome measurement
62	Equity / Bias Monitoring (detect uneven model performance)	✗ Not Started	0%	No	NEEDED: cohort-level analytics overlay	No bias detection
63	Wellbeing Safeguards (child protection + human support boundaries)	✗ Not Started	0%	No	NEEDED: escalation policy + safeguard layer	Wellbeing CRUD exists; no safeguard framework
64	AI / Data Ethics Framework (child-centered, beyond RBAC)	◆ Partial	20%	No	JWT + RBAC + tenant isolation done	No child-ethics policy or consent framework
— 11. FUTURE FEATURES (Architecture doc roadmap) —						
65	Voice Tutor (speech-to-text + AI response + TTS)	✗ Not Started	0%	No	NEEDED: extend speech module	Speech transcription exists; full voice loop not wired
66	Image Understanding (student uploads photo of problem)	✗ Not Started	0%	No	NEEDED: vision model integration (LLaVA/Qwen-VL)	No vision model integrated
67	Video Understanding (lecture video → summary + quiz)	✗ Not Started	0%	No	NEEDED: video ingestion pipeline	No video processing
68	Code Tutor (programming support + code review)	✗ Not Started	0%	No	NEEDED: code-specific prompt + execution sandbox	No code tutor mode
69	Math Solver (step-by-step equation solving)	✗ Not Started	0%	No	NEEDED: math-aware model or tool (Qwen-Math)	No math-specific model
70	Topic Failure Prediction (predict fail before test happens)	✗ Not Started	0%	No	NEEDED: time-series model over mastery decay	Requires longitudinal mastery data
71	Learning Style Detection (visual/reading/quiz preference)	✗ Not Started	0%	No	NEEDED: interaction log clustering	No interaction preference tracking
72	Performance Forecasting (predicted score on next test)	✗ Not Started	0%	Yes	NEEDED: regression over mastery + engagement data	Listed in old CSV as Future ML
73	At-risk / Dropout Risk Prediction (ML model)	✗ Not Started	0%	Yes	NEEDED: attendance + mastery + engagement pipeline	Listed in old CSV as Future ML

EEC ML — Features Missing from Old CSV Checklist				
Feature	Status	Progress %	Files / Location	Why It Was Missing
— ALREADY BUILT — but not in the old checklist				
Embedding Engine	✅ Done	95%	ai-service/app/modules/embeddings/	Standalone engine; listed within ingestion only
Explain Mode (AI Tutor)	✅ Done	90%	ai-service chat module (mode=explain)	One of 7 modes; not listed separately
ML Engine (EMA mastery)	🟡 Partial	50%	backend/services/mlEngine.js	Weighted mastery + at-risk + class trends; completely absent
Engagement Scorer	🟡 Partial	50%	backend/services/engagementScorer.js	Topic-level engagement score; completely absent
Spaced Repetition (SM-2)	🟡 Partial	55%	backend/models/SpacedRepetitionSchedule.js	SM-2 algorithm built; listed in old CSV as 'Future ML' — wrong
Recommendation Engine	🟡 Partial	40%	backend/routes/recommendationRoutes.js	Route with mastery+spaced-rep logic; completely absent
At-risk Student Detection	🟡 Partial	40%	backend/routes/mlRoutes.js (GET /class/at-risk)	Endpoint live; not in old CSV
Engagement Analytics	🟡 Partial	40%	backend/routes/engagementRoutes.js	Endpoint live; not in old CSV
AI Lesson Content (Teacher)	🟡 Partial	45%	backend/routes/aiTeacherRoutes.js (lesson-content)	Teacher AI feature; completely absent
AI Hinge Question (Teacher)	🟡 Partial	45%	backend/routes/aiTeacherRoutes.js (hinge-questions)	Teacher AI feature; completely absent
Wellbeing Tracking	🟡 Partial	40%	backend/models/Wellbeing.js + wellbeingRoute.js	Model + CRUD exist; completely absent
Answer Evaluator (full)	🟡 Partial	40%	ai-service/app/modules/assessment/	Full 10-field evaluator; listed as 'Assessment Engine' only
— ARCHITECTURAL — not in old checklist				
AI Orchestrator	❌ Not Started	0%	NEEDED: ai-service/app/orchestrator/	Central coordinator; Node calls endpoints directly
Prompt Library	❌ Not Started	0%	NEEDED: ai-service/prompts/	All prompts hardcoded — arch violation
Hybrid Search (BM25+vector)	❌ Not Started	0%	NEEDED in retrieval/service.py	Only vector search today
Document Versioning	❌ Not Started	0%	NEEDED in TeachingMaterial.js	Arch doc mandates: never auto-delete old versions
Learning Outcomes Auto-gen	❌ Not Started	0%	NEEDED in documents/service.py	Key step in ingestion pipeline — missing
Bloom Engine	❌ Not Started	0%	NEEDED: ai-service/app/modules/bloom/	Zero code; doc claims ✅ Strong (incorrect)
— CONTENT GENERATION — not in old checklist				
Worksheet / Homework Gen	❌ Not Started	0%	NEEDED in chat module	No personalised worksheet output
Adaptive Difficulty	❌ Not Started	0%	NEEDED: wire MasteryScore → content generation	Fixed difficulty today
AI Question Paper (Teacher)	❌ Not Started	0%	NEEDED: Bloom + difficulty param in exam gen	No Bloom-level selection for teachers
Teacher AI Override	❌ Not Started	0%	NEEDED: annotation layer on AI responses	No mechanism to flag/correct AI answers
— RESEARCH-GRADE — not in old checklist				
Learner Confidence	❌ Not Started	0%	NEEDED: new model + AI check-in flow	No data collection
Help-Seeking Behaviour	❌ Not Started	0%	NEEDED: event logging in tutor session	No tracking
Misconceptions Engine	❌ Not Started	0%	NEEDED: extend error classifier	Error classifier itself not built
Multidimensional Engagement	🟡 Partial	25%	engagementScorer.js (behavioural only)	Situational + emotional dimensions missing
Intervention Effectiveness	❌ Not Started	0%	NEEDED: analytics on interventions	InterventionLog.js exists; no measurement
Teacher Escalation	❌ Not Started	0%	NEEDED: alert from gap/at-risk engine	No escalation workflow
Student Agency	❌ Not Started	0%	NEEDED: path selection UI	All paths teacher-assigned
Explainable AI	❌ Not Started	0%	NEEDED: reasoning trace in tutor response	Black box
Social / Belonging	❌ Not Started	0%	Alcove posts exist; no belonging analytics	No social learning signals
Evidence Testing	❌ Not Started	0%	NEEDED: pre/post outcome measurement	No framework
Equity / Bias Monitoring	❌ Not Started	0%	NEEDED: cohort analytics overlay	No bias detection
Wellbeing Safeguards	❌ Not Started	0%	NEEDED: escalation policy + child protection	Wellbeing CRUD exists; no safeguard layer
AI / Data Ethics Framework	🟡 Partial	20%	JWT + RBAC + tenant isolation done	Child-ethics policy not designed
— FUTURE — not in old checklist				
Voice Tutor	❌ Not Started	0%	Extend speech module	Transcription exists; full voice loop not wired
Image Understanding	❌ Not Started	0%	NEEDED: vision model (LLaVA/Qwen-VL)	No vision model
Video Understanding	❌ Not Started	0%	NEEDED: video ingestion pipeline	No video processing
Code Tutor	❌ Not Started	0%	NEEDED: code-mode prompt + sandbox	No code tutor mode
Math Solver	❌ Not Started	0%	NEEDED: math model (Qwen-Math)	No math-specific model
Topic Failure Prediction	❌ Not Started	0%	NEEDED: time-series over mastery decay	Not in old CSV at all
Learning Style Detection	❌ Not Started	0%	NEEDED: interaction log clustering	Not in old CSV at all

EEC ML — Backend Models					
Model File	Purpose	Status	Progress %	Route File	Notes / Gaps
— AUTH & USERS —					
Admin.js	Admin user schema	✅ Done	100%	adminRoutes.js	Full CRUD + RBAC
ParentUser.js	Parent account	✅ Done	100%	parentRoute.js	Full portal
Principal.js	Principal account	✅ Done	100%	principalRoutes.js	Full portal
StaffUser.js	Non-teaching staff	✅ Done	100%	staffRoutes.js	Full portal
StudentUser.js	Student account	✅ Done	100%	studentRoute.js	Full portal
TeacherUser.js	Teacher account	✅ Done	100%	teacherRoute.js	Full portal
SuperAdminActivity.js	Super-admin audit log	✅ Done	100%	superAdminRoutes.js	Audit trail
— SCHOOL STRUCTURE —					
Organization.js	Top-level tenant	✅ Done	100%	organizationRoutes.js	Multi-tenant root
School.js	School entity	✅ Done	100%	schoolRoutes.js	Full CRUD
AcademicYear.js	Academic year	✅ Done	100%	academicRoutes.js	Full CRUD
Class.js	Class / grade	✅ Done	100%	academicRoutes.js	Full CRUD
Section.js	Section in class	✅ Done	100%	academicRoutes.js	Full CRUD
Subject.js	Subject entity	✅ Done	100%	subjectRoute.js	Full CRUD
Department.js	Teacher department	✅ Done	100%	departmentRoutes.js	Full CRUD
Timetable.js	Class timetable	✅ Done	100%	timetableRoutes.js	Full CRUD
Building.js	Campus building	✅ Done	100%	—	Reference data
Floor.js	Floor in building	✅ Done	100%	—	Reference data
Room.js	Classroom entity	✅ Done	100%	—	Reference data
Holiday.js	Holiday calendar	✅ Done	100%	holidayRoutes.js	Full CRUD
TeacherAllocation.js	Teacher → class-subject	✅ Done	100%	teacherAllocationRoutes.js	Full CRUD
— ACADEMIC & EXAMS —					
Assignment.js	Assignment + submissions	✅ Done	100%	assignmentRoute.js	Full CRUD
Exam.js	Exam entity	✅ Done	100%	examRoute.js	Full CRUD
ExamAttempt.js	Student exam attempt	✅ Done	100%	examRoute.js	Recorded
ExamGroup.js	Exam grouping	✅ Done	100%	examRoute.js	Full CRUD
ExamQuestion.js	MCQ/short/long question	⚠️ Partial	50%	examRoute.js	No Bloom level; no difficulty calibration
ExamResult.js	Exam result record	✅ Done	100%	examRoute.js	Full CRUD
LessonPlan.js	Teacher lesson plan	✅ Done	100%	lessonPlanRoutes.js	Full CRUD
LessonPlanCompletion.js	Lesson completion track	✅ Done	100%	lessonPlanRoutes.js	Tracked
ReportCardTemplate.js	Report card template	✅ Done	100%	reportRoutes.js	Full CRUD
TeachingMaterial.js	Teacher-uploaded material	✅ Done	100%	teachingMaterialRoutes.js	Cloudinary + Qdrant ingest
Rubric.js	Grading rubric	⚠️ Partial	60%	rubricRoutes.js	Not linked to answer evaluator
PromotionHistory.js	Student promotion record	✅ Done	100%	promotionRoutes.js	Full CRUD
— AI & LEARNING —					
CurriculumMap.js	Topic prerequisite graph	⚠️ Partial	30%	curriculumMapRoutes.js	No traversal; no auto-update on ingest
MasteryScore.js	Per-topic mastery score	⚠️ Partial	60%	masteryRoutes.js	Not auto-updated after quiz/test
BaselineQuiz.js	Baseline quiz schema	✅ Done	85%	baselineRoutes.js	Results not feeding mastery
BaselineResult.js	Baseline quiz result	⚠️ Partial	60%	baselineRoutes.js	Not wired to mastery update
PracticePaper.js	Practice paper	✅ Done	100%	practicePaperRoutes.js	Full CRUD
PracticeQuestion.js	Practice question	⚠️ Partial	50%	practiceRoutes.js	No Bloom; not AI-generated
PracticeSection.js	Practice paper section	✅ Done	100%	practiceSectionRoutes.js	Full CRUD
PracticeAttempt.js	Practice attempt	⚠️ Partial	60%	practiceRoutes.js	Not feeding mastery
TryoutResult.js	AI tryout result	✅ Done	90%	aiLearningRoute.js	Ratings not feeding mastery
FlashcardResult.js	Flashcard rating result	⚠️ Partial	40%	—	Model exists; not persisted from UI
ExternalResource.js	External study links	✅ Done	100%	externalResourceRoutes.js	Full CRUD
SpacedRepetitionSchedule.js	SM-2 review schedule	⚠️ Partial	55%	spacedRepetitionRoutes.js	SM-2 logic built; not auto-triggered from results
TeacherLearningPath.js	Teacher learning path	⚠️ Partial	45%	learningPathRoutes.js	Manual only; no AI generation
StudentProgress.js	Student progress snapshot	⚠️ Partial	50%	progressRoute.js	Not linked to mastery/gap
StudentMemorySummary.js	AI student memory summary	⚠️ Partial	30%	—	Academic memory pipeline partial
TutorConversation.js	AI tutor chat session	⚠️ Partial	40%	aiTutorRoutes.js	Per-session only; no cross-session recall
ReadingMaterial.js	Reading passage upload	✅ Done	100%	readingAssessmentRoutes.js	Full CRUD
ReadingAssessment.js	Reading attempt + scores	✅ Done	95%	readingAssessmentRoutes.js	End-to-end pipeline working
WritingPrompt.js	Writing prompt upload	✅ Done	100%	writingAssessmentRoutes.js	Full CRUD
WritingAssessment.js	Writing submission + scores	✅ Done	95%	writingAssessmentRoutes.js	End-to-end pipeline working
StudentLanguageProfile.js	Adaptive language profile	✅ Done	85%	—	Qdrant memory active
— STUDENT WELFARE —					
StudentBadge.js	Achievement badge	⚠️ Partial	60%	achievementRoutes.js	Badge model + masteryEngine award logic; no full auto-a
StudentJournalEntry.js	Personal journal	⚠️ Partial	40%	—	Model exists; no UI
StudentObservation.js	Teacher observation note	⚠️ Partial	50%	studentObservationRoutes.js	Not linked to intervention
Wellbeing.js	Wellbeing check-in	⚠️ Partial	40%	wellbeingRoute.js	No safeguard/escalation layer
InterventionLog.js	Teacher intervention record	⚠️ Partial	30%	—	No auto-trigger from gaps
Behaviour.js	Behaviour incident	✅ Done	100%	behaviourRoute.js	Full CRUD
ExcuseLetter.js	Excuse letter submission	✅ Done	100%	excuseLetterRoutes.js	Full CRUD
— COMMUNICATION & FINANCE —					
ChatThread.js	Real-time chat thread	✅ Done	100%	chatRoutes.js	Socket.IO active
ChatMessage.js	Chat message	✅ Done	100%	chatRoutes.js	Active
Notification.js	Push notification	✅ Done	100%	notificationRoutes.js	Web push + in-app
PushSubscription.js	Browser push sub	✅ Done	100%	notificationRoutes.js	VAPID keys active
AlcovePost.js	Academic Alcove post	✅ Done	100%	alcoveRoute.js	Full CRUD
AlcoveComment.js	Alcove comment	✅ Done	100%	alcoveRoute.js	Full CRUD
AlcoveSubmission.js	Alcove submission	✅ Done	100%	alcoveRoute.js	Full CRUD
FeeStructure.js	Fee structure	✅ Done	100%	feeRoutes.js	Full CRUD
FeeInvoice.js	Fee invoice	✅ Done	100%	feeRoutes.js	Full CRUD
FeePayment.js	Razorpay payment	✅ Done	100%	feeRoutes.js	Razorpay integrated
PaymentAudit.js	Payment audit trail	✅ Done	100%	paymentSettingsRoutes.js	Full audit log
AuditLog.js	System audit log	✅ Done	100%	auditLogRoutes.js	All key events logged

EEC ML — AI Service Modules

Module	Responsibility	What It Does	Status	Progress %	Notes / Gaps
— BUILT —					
documents	Document ingestion	OCR → parse → chunk → embed → Qdrant upsert	✅ Done	90%	Topic auto-detect + Bloom tagging missing
parser	Text extract + cleaner	PyMuPDF/Tesseract/docx/pptx; strip teacher notes	✅ Done	90%	_strip_teacher_notes() active
embeddings	Embedding generation	nomic-embed-text → Qdrant	✅ Done	95%	Fully wired
retrieval	Semantic retrieval	Qdrant metadata filters: school/class/section/subject/chapter	✅ Done	90%	Hybrid BM25+semantic not yet done
chat	RAG tutor — all modes	explain/quiz/flashcards/notes/mindmap/summarize/homework	✅ Done	90%	Prompts hardcoded; no prompt library
quiz	Quiz generation	5 MCQ per request via RAG	✅ Done	80%	No Bloom; not saved to question bank
flashcards	Flashcard generation	Q:/A: format via RAG	✅ Done	85%	Ratings not persisted to mastery
summaries	Summary generation	Chapter summaries; OCR + summarize endpoint	✅ Done	90%	Working well
assessment	Language eval (Qwen3 8B)	Reading + writing rubric + score JSON	✅ Done	90%	Academic MCQ eval not wired
speech	Audio transcription	faster-whisper + SpeechBrain pronunciation	✅ Done	85%	Real-device latency testing needed
language_memory	Adaptive language memory	Qdrant student_language_memory; store/retrieve	✅ Done	80%	Academic memory not included
admin	Admin module	Admin-facing AI controls	🟡 Partial	20%	Scaffolded; not fully implemented
— NOT YET BUILT —					
orchestrator	Central AI coordinator	Node → Orchestrator → engines	❌ Not Started	0%	Biggest arch violation
knowledge_graph	Graph traversal	Walk prerequisites; find root-cause weak topics	❌ Not Started	0%	Model in Node; no Python engine
bloom	Bloom classifier	Tag documents, questions, answers	❌ Not Started	0%	Not in any pipeline step
mastery_engine	Mastery calculator	accuracy + attempts + time + recency; auto-update	❌ Not Started	0%	masteryEngine.js in Node; no Python module
error_classifier	Error type classifier	Concept/Calculation/Reading/Logic	❌ Not Started	0%	No service, no route, no model
gap_detector	Gap detection	Prerequisite traversal → root-cause weak topic	❌ Not Started	0%	Blocked by error classifier
recommendation	Recommendation engine	Next topic based on mastery + curriculum graph	❌ Not Started	0%	Node route exists; no AI engine
analytics	AI analytics insights	Per-topic heatmap; at-risk; AI narrative	❌ Not Started	0%	Basic Node routes exist
prompt_library	Prompt directory	externalise all prompts from Python modules	❌ Not Started	0%	Arch violation — all prompts hardcoded

EEC ML — Master Progress Summary (2026-08-11)

Domain	Total Features	✔ Done	⬢ Partial	✖ Not Started	Next Action
Core AI Pipeline	6	3	0	3	Build Orchestrator + Prompt Library first
Content Intelligence	5	0	2	3	Bloom Engine + Learning Outcomes + Doc Versioning
Assessment & Evaluation	6	3	1	2	Wire academic MCQ eval + full Answer Evaluator
Mastery & Memory	6	1	4	1	Auto-update mastery + build academic memory
Gap & Learning Path	4	0	0	4	Error Classifier → Gap Detector → AI Path
AI Content Generation (Student)	9	7	0	2	Adaptive Difficulty + Worksheet Generation
AI Content Generation (Teacher)	4	0	2	2	Bloom param for question paper + Teacher Override
Language Assessment	3	3	0	0	Complete — minor polish only
Analytics & Intelligence	7	0	5	2	Surface at-risk + engagement into teacher UI
Research-Grade Features	14	0	3	11	Post-foundation — requires mastery+gap pipeline first
Future Features	9	0	0	9	Architecture-ready; build after core is solid
TOTAL	73	17	17	39	

Overall: 17 Done + 17 Partial out of 73 total features (23% fully done)

OLD CSV HAD 31 ITEMS — ACTUAL TOTAL IS 73 AI FEATURES

42 features were completely missing from the old checklist (see 'Missing from Old CSV' sheet)

TOP 5 PRIORITIES

1. Error Classification Engine	Nothing downstream works without it — blocks Gap Detection + Intervention
2. Mastery Engine Auto-Update	Wire quiz/test/assessment results → MasteryScore after every attempt
3. Gap Detection Engine	Core intelligence: traverse curriculum graph → root-cause weak topics
4. AI Orchestrator	Fix arch violation — Node must call Orchestrator, not individual endpoints
5. Prompt Library	Externalise all hardcoded prompts — required by architecture doc